3.1.2 Amount of substance: Sheet 1

Moles, empirical formula and reacting masses · 17 marks · about 25 minutes

Key information

- **$n = m \div M_r$** and **$n = c \times V$** (V in dm^3: $cm^3 \div 1000$).
- Empirical formula: divide each mass (or %) by A_r, then divide by the smallest; round to whole numbers.
- Water of crystallisation: mass of water lost = mass before heating − mass after heating; ratio of moles salt : moles H_2O gives x.
- Reacting masses: moles of known → use the equation ratio → moles of unknown → mass.
- Limiting reactant: work out moles of each reactant and use the equation ratio to see which runs out first.

Section A: Self-quiz

01 What is the empirical formula of an oxide of chlorine that contains 42.5% by mass of chlorine?

[1 mark]

A ClO_2
B ClO_3
C Cl_2O_3
D Cl_2O_5

02 Which of these contains the greatest number of atoms?

[1 mark]

A 127 mg of iodine
B 1.54×10^{-4} kg of phosphorus
C 81.0 mg of carbon dioxide
D 1.70×10^{-4} kg of ammonia

03 A 4.85 g sample of anhydrous sodium sulfate is dissolved in water and the solution made up to 250 cm^3 in a volumetric flask. What is the concentration in mol dm^{-3} of sodium sulfate in the solution?

[1 mark]

A 0.0341
B 0.137
C 0.163
D 0.273

04 What is the maximum mass of sodium chloride that can be made from 10.0 g of sodium and 10.0 g of chlorine?

[1 mark]

A 8.24 g
B 16.5 g
C 20.0 g
D 25.4 g

05 What is the empirical formula of an oxide of nitrogen that contains 26% nitrogen by mass?

[1 mark]

A NO_2
B N_2O_3
C N_2O_5
D N_4O_5

Section B: Exam questions

06 A student heated a solid sample of $Na_2CO_3.xH_2O$ for 1 minute to remove water and determine a value for x

Table 4 shows the results recorded.

Table 4

Mass of empty evaporating basin	24.35 g
Mass of evaporating basin and solid before heating	25.47 g
Mass of evaporating basin and solid after heating for 1 minute	24.92 g

06.1 Use the data in Table 4 to calculate a value for x in the formula $Na_2CO_3.xH_2O$

Give your answer to 2 decimal places.

[5 marks]

06.2 The correct value for x is 10

Suggest a reason for the difference between the experimental value for x and the correct value.

(If you were unable to calculate an experimental value for x assume it was 8.05. This is not the correct experimental value.)

[1 mark]

06.3 Suggest how the procedure could be improved, using the same apparatus, to give a more accurate value for x

Justify your answer.

[2 marks]

07 A mixture of magnesium oxide and magnesium hydroxide has a mass of 3200 mg

This mixture is reacted with carbon dioxide to form magnesium carbonate and water.

The mass of water produced is 210 mg

$Mg(OH)_2 + CO_2 \rightarrow MgCO_3 + H_2O$

$MgO + CO_2 \rightarrow MgCO_3$

Calculate the percentage by mass of magnesium oxide in this mixture.

[4 marks]

Mark Scheme

M1, M2… are separate marking points. / separates acceptable alternatives. Words in brackets are not needed for the mark. ECF = error carried forward. Questions and mark schemes © AQA, reproduced for internal school use.

Question	Marking guidance	Comments	Mark
01	**B**	Source: 7404/1 Jun 2024 Q18 ClO_3	1
02	**D**	Source: 7404/1 Jun 2017 Q15	1
03	**B**	Source: 7404/1 Jun 2017 Q14	1
04	**B**	Source: 7404/1 Jun 2025 Q13 16.5 g	1
05	**C**	Source: 7404/1 Jun 2017 Q22	1
06.1	M1 Mass Na_2CO_3 = 0.57 g AND mass H_2O = 0.55 g M2 Mol Na_2CO_3 = 0.57 ÷ 106 AND mol H_2O = 0.55 ÷ 18 M3 = 0.0054 : 0.0306 M4 ÷ by smallest = 1 : 5.682 M5 Value of x = 5.68 (2 dp)	Source: 7404/1 Jun 2018 Q03.1–03.3 If incorrect masses other than AE, lose M1 & M3 M3 = these values only (at least 2 sf) M5 allow 5.67 – 5.74 OR: M1 mass Na_2CO_3 = 0.57 g AND mass $Na_2CO_3.xH_2O$ = 1.12 g; M2 moles $Na_2CO_3 = 5.377 \times 10^{-3}$; M3 M_r hydrated = 208.3; M4 M_r of xH_2O = 102.3; M5 x = 5.68	5
06.2	Failure to drive off all the water / failure to heat for long enough / not heated to constant mass	Allow evaporate instead of drive off Ignore incomplete reaction	1
06.3	M1 Heat to constant mass / heat for longer / use a smaller mass M2 You can be sure all / more of the water has been driven off	Ignore incomplete reaction M2 dependent on M1	2
07	M1 moles of water in 210 mg = 0.210 ÷ 18 = 0.0117 mol (equal to moles of magnesium hydroxide) M2 mass of $Mg(OH)_2$ = 0.0117 × 58.3 = 0.680 g M3 mass of MgO = 3.2 – 0.68 = 2.52 g M4 % of MgO = 2.52 ÷ 3.2 × 100 = 78.7%	Source: 7404/1 Jun 2019 Q07.2 Allow correct alternative methods, eg M3 = % $Mg(OH)_2$ = M2 × 100 ÷ 3.2, then M4 = 100 – M3 M4 allow 78.7 – 78.8 or 79%	4